

20. (b) 21. (d) 22. (d) 23. (b) 24. (b) 25. (b) 26. (a) 27. (a) 28. (a) 29. (c)
 30. (a) 31. (c) 32. (b) 33. (c) 34. (a) 35. (c) 36. (c) 37. (c) 38. (d) 39. (c)
 40. (a) 41. (b) 42. (c) 43. (c) 44. (a) 45. (a) 46. (b) 47. (c) 48. (c) 49. (c)
 50. (a)

Solutions with Necessary Explanation

1. $468 \div 39 = 12$

Therefore, 11 numbers are available and the number of pair of numbers having HCF = 39 is given by $\frac{11+1}{2} = 6$

2. (i) The least number which is divisible by 6, 8 and 12 is the LCM of 6, 8 and 12 = 24.

The numbers greater than 500 and less than 700 which are divisible by 24 are

$$504, 528, 552, \dots, 696.$$

$$\begin{aligned}\text{The sum of the numbers} &= \frac{9}{2} (504 + 696) \\ &= 9 \times 600 = 5400\end{aligned}$$

- (ii) The number leaves remainder as 5 in each case, the required numbers are greater by 5 than the above numbers

$$\begin{aligned}\text{i.e., } &504 + 5, 528 + 5, \dots, 696 + 5 - 24 \\ &= 509, 533, \dots, 677.\end{aligned}$$

$$\begin{aligned}\text{The sum of the numbers} &= \frac{8}{2} (509 + 677) \\ &= 4 \times 1186 \\ &= 4744\end{aligned}$$

3. Since 1000 is not divisible by 15, it is not possible to divide 1000 into two parts such that their HCF may be 15. Hence the correct answer is (d).

4. Number of boys = 442; Number of girls = 374

As the number of boys in the class is same as that of girls.

$$\text{Strength of one class} = \text{HCF of } 442 \text{ and } 374 = 34$$

$$\therefore \text{Number of classes of boys} = 442 \div 34 = 13$$

$$\text{Number of classes of girls} = 374 \div 34 = 11$$

$$\therefore \text{Required number of classes} = 13 + 11 = 24$$

5. The greatest number of 4 digits = 9999

On dividing 9999 by 196, 3 will be left as remainder.

$$\therefore \text{The required greatest number} = 9996.$$

$$\text{Least number of 5 digits} = 10000$$

On dividing 10000 by 196, 4 will be left as remainder.

$$\therefore \text{The required 5 digit number} = 10000 - 4 + 196 = 10192$$

6. Let x and y be the two numbers and their HCF = 9; product = 4212.

$$\therefore x \times y = 9^2 \quad (\text{product in lower terms})$$

$$\text{i.e. } 4212 = 81 \quad (\text{product in lower terms})$$

$$\text{Product in lower terms} = 52$$

The possible values are (1, 52) or (4, 13).

∴ The required numbers are 4×9 and 13×9 , i.e. **36 and 117**.

7. The given HCF is 36 and LCM is 756.

$$\begin{aligned}\therefore \text{Product of two numbers} &= \text{HCF} \times \text{LCM} \\ &= 36 \times 756\end{aligned}$$

$$(\text{Product in lower terms}) \times (\text{HCF})^2 = \text{HCF} \times \text{LCM}$$

$$\text{Product in lower terms} = \frac{\text{LCM}}{\text{HCF}} = \frac{756}{36} = 21$$

The numbers are multiples of 3 and 7.

∴ The required numbers are 3×36 and 7×36 , i.e. **108, 252**.

8. Required maximum length = HCF of 6 m, 4 m 75 cm and 10 m 25 cm
i.e. the HCF of 600 cm, 475 cm and 1025 cm = **25 cm**

9. Let the second number be x .

$$x \times \text{the first number} = \text{HCF} \times \text{LCM}$$

$$\therefore x \times 144 = 36 \times 1008$$

$$\therefore x = \frac{36 \times 1008}{144} = 252$$

∴ Required number = **252**

10. Since the bells ring at the intervals of 6 seconds, 7 seconds and 8 seconds, respectively, next time they will ring together after an interval equal to the LCM of 6, 7 and 8 seconds.

$$\begin{aligned}\text{Therefore, the required time} &= \text{LCM of 6, 7 and 8 seconds} \\ &= \mathbf{168 \text{ seconds}}\end{aligned}$$

11. The maximum size of square tile
= HCF of length and breadth of hall
= HCF of 17 m 55 cm and 8 m 10 cm
= HCF of 1755 cm and 810 cm
= 135 cm

$$\text{The number of square tiles required} = \frac{\text{Area of hall}}{\text{Area of one tile}} = \frac{1755 \times 810}{(135)^2} = \mathbf{78}$$

12. Required number = LCM of 6, 8, 15, 18 and 24
= **360**

13. We have

$$36 = 2 \times 2 \times 3 \times 3$$

$$84 = 2 \times 2 \times 3 \times 7$$

$$\therefore \text{HCF} = 2 \times 2 \times 3 = \mathbf{12}$$

14.

2	24,	36,	40
2	12,	18,	20
2	6,	9,	10
3	3,	9,	5
	1,	3,	5

$$\therefore \text{LCM} = 2 \times 2 \times 2 \times 3 \times 3 \times 5 = \mathbf{360}$$

15. Let the numbers be $3x$ and $4x$.

HCF of above two numbers = x (given as 4)

$\therefore$ Number are 12 and 16.

LCM of 12 and 16 = **48**.

16. Let the numbers be $37x$ and $37y$.

$$\therefore 37x \times 37y = 4107$$

$$\Rightarrow x \times y = \frac{4107}{37 \times 37} = 3$$

$$\text{i.e. } x \times y = 1 \times 3 = 3$$

$$\therefore \text{Greater number} = 37y = 37 \times 3 = \mathbf{111}$$

17. Given HCF = 21; LCM = 84;

Let the numbers be $21x$ and $21y$.

$$\therefore \frac{21x}{21y} = \frac{1}{4}$$

$$\text{or } \frac{x}{y} = \frac{1}{4}$$

$$\therefore \text{Larger number} = 21 \times 4 = \mathbf{84}$$

18. Given HCF of two numbers = 8

LCM is completely divisible by HCF

$\therefore$ 24, 48 and 56 are completely divisible by 8, the value which can never be LCM = **60**.

19. Let the numbers be $29x$ and $29y$

$$\therefore \text{LCM} = 29 \times x \times y = 4147$$

$$\Rightarrow xy = \frac{4147}{29} = 143$$

Values of x and y can be (1, 143) and (11, 13).

Given that both numbers are greater than 29.

$\therefore$ Required numbers are 29×11 , 29×13 . i.e. 319, 377

$\therefore$ Sum of numbers = $319 + 377 = \mathbf{696}$

20. LCM of 5, 6, 7 and 8

$$\begin{array}{c|cccc} 2 & 5, & 6, & 7, & 8 \\ \hline & 5, & 3, & 7, & 4 \end{array}$$

$$= 2 \times 5 \times 3 \times 7 \times 4 = \mathbf{840}$$

$\therefore$ The required number = $840k + 3$.

For the number to be divisible by 9, $k = 2$

i.e., $840 \times 2 + 3 = \mathbf{1683}$

21. LCM of 6, 9, 15 and 18

$$\begin{array}{c|cccc} 2 & 6, & 9, & 15, & 18 \\ \hline 3 & 3, & 9, & 15, & 9 \\ \hline 3 & 1, & 3, & 5, & 3 \\ \hline & 1, & 1, & 5, & 1 \end{array}$$

$$= 2 \times 3 \times 3 \times 1 \times 1 \times 5 \times 1 = \mathbf{90}$$

Let the number be $90k + 4$.

$\therefore$ Number is divisible by 7, least value of $k = 4$.

$\therefore$ Required number = $90 \times 4 + 4 = 364$

22. LCM of 12, 15, 20 and 54

2	12,	15,	20,	54
2	6,	15,	10,	27
3	3,	15,	5,	27
5	1,	5,	5,	9
3	1,	1,	1,	9
	1,	1,	1,	3

$$= 2 \times 2 \times 3 \times 3 \times 3 \times 5 = 540$$

$\therefore$ Least number which leaves remainder 8 = $540 + 8 = 548$

23. LCM of 12, 16, 18, 21 and 28

2	12,	16,	18,	21,	28
2	6,	8,	9,	21,	14
3	3,	4,	9,	21,	7
7	1,	4,	3,	7,	7
2	1,	4,	3,	1,	1
	1,	2,	3,	1,	1

$$= 2 \times 2 \times 3 \times 7 \times 2 \times 2 \times 3 = 2^4 \times 3^2 \times 7 = 1008$$

$\therefore$ Smallest number diminished by 7 which is divisible by 12, 16, 18, 21 and 28

$$= 1008 + 7 = 1015$$

24. LCM of 12, 18, 21 and 30

2	12,	18,	21,	30
2	6,	9,	21,	15
3	3,	9,	21,	15
	1,	3,	7,	5

$$= 2 \times 2 \times 3 \times 3 \times 5 \times 7 = 1260$$

$$\therefore \text{Required number} = \frac{1260}{2} = 630$$

25. HCF of $(1657 - 6)$ and $(2057 - 5) = \text{HCF of } 1651 \text{ and } 2032$

$$\begin{array}{r}
 1651 \overline{) 2032} \quad (1 \\
 \underline{1651} \\
 381 \\
 381 \overline{) 1651} \quad (4 \\
 \underline{1524} \\
 127 \\
 127 \overline{) 381} \quad (3 \\
 \underline{381} \\
 0
 \end{array}$$

$\therefore \text{HCF} = 127$

26. $N = \text{HCF of } (6905 - 4665), (4665 - 1305) \text{ and } (6905 - 1305)$

i.e. $N = \text{HCF of } 2240, 3360 \text{ and } 5600 = 1120$

$\therefore$ Sum of digits of $N = 1 + 1 + 2 + 0 = 4$

27. Required number = HCF of $(91 - 43), (183 - 91) \text{ and } (183 - 43)$

= HCF of 48, 92 and 140 = 4

28. HCF of 1001 and 910 = 91

$$\begin{array}{r} 910 \overline{) 1001} \quad (1 \\ \underline{910} \\ 910 \quad (10 \\ \underline{910} \\ 0 \end{array}$$

$\therefore$ Required number of students = 91

29. Required length = HCF of 7 m, 3 m 85 cm and 12 m 95 cm

i.e. HCF of 700 cm, 385 cm, and 1295 cm

$$\begin{array}{r} 385 \overline{) 700} \quad (1 \\ \underline{385} \\ 315 \quad (1 \\ \underline{315} \\ 0 \end{array}$$

HCF = 35

HCF of 1295 and 35 = 35

$$\begin{array}{r} 35 \overline{) 1295} \quad (37 \\ \underline{705} \\ 245 \\ \underline{245} \\ 0 \end{array}$$

30. Given HCF = 11; LCM = 385;

$$385 = 5 \times 7 \times 11$$

$\therefore$ Number lies between 75 and 125 = $7 \times 11 = 77$

31. Given HCF = 23; LCM = $13 \times 14 \times 23$;

$\therefore$ Larger of the two numbers = $14 \times 23 = 322$

32. HCF of co-prime numbers = 1

Product of co-prime numbers = 117

$$\therefore \text{LCM of co-prime numbers} = \frac{117}{1} = 117$$

33. Let the numbers be x and y .

$$\text{Given, } x + y = 55 \quad (1)$$

$$\text{HCF} = 5; \text{LCM} = 120;$$

$$\text{We have } \text{HCF} \times \text{LCM} = x \times y$$

$$\text{i.e. } 5 \times 120 = xy \quad (2)$$

$$\begin{aligned} \text{i.e.} \quad x + y &= 55 & (1) \\ xy &= 600 & (2) \end{aligned}$$

$$\therefore \frac{1}{x} + \frac{1}{y} = \frac{y+x}{xy} = \frac{55}{600} = \frac{11}{120}$$

34. Let the numbers be x and y .

$$\text{Given} \quad x + y = 100 \quad (1)$$

LCM = 495; HCF = 5;

$$\therefore \text{LCM} \times \text{HCF} = xy$$

$$\text{i.e.,} \quad 495 \times 5 = xy$$

$$\text{or} \quad xy = 495 \times 5 \quad (2)$$

$$x - y = ?$$

$$\begin{aligned} (x - y)^2 &= (x + y)^2 - 4xy \\ &= (100)^2 - 4 \times 495 \times 5 \\ &= (100)^2 - 4 \times 5^2 \times 99 \\ &= 10^2 (10^2 - 99) = 100 \end{aligned}$$

$$\therefore x - y = 10 \quad (3)$$

$\therefore$ Difference between numbers = 10

35. Given HCF = 11; LCM = 7700; $x = 275$; $y = ?$

$$\text{We have,} \quad x \times y = \text{HCF} \times \text{LCM}$$

$$\text{i.e.,} \quad 275 \times y = 11 \times 7700$$

$$\therefore y = \frac{11 \times 7700}{275} = \frac{7700}{25} = 77 \times 4 = \mathbf{308}$$

36. Given $x \times y = 2028$; HCF = 13;

$$\therefore \text{LCM} = \frac{x \times y}{\text{HCF}} = \frac{2028}{13} = 156$$

Let $x = 13a$ and $y = 13b$

$$\therefore x \times y = 13 \times 13 \ ab = 13 \times 156 = 13 \times 13 \times 12$$

$$\therefore ab = 12$$

Factors of 12 are (1, 12), (2, 6) and (3, 4).

There are **3** such pairs of numbers.

37. Let the numbers be $33a$ and $33b$.

$$\therefore 33a + 33b = 528$$

$$\text{i.e.,} \quad a + b = \frac{528}{33} = 16$$

$\therefore$ Values of (a, b) can be (1, 15), (2, 14), (3, 13), (4, 12), (5, 11), (6, 10), (7, 9) and (8, 8)

$\therefore$ Number of pairs of numbers satisfying the given condition = 8

38. Given ratio of numbers = 1 : 2 : 3;

Their HCF = 12

$\therefore$ Numbers are **12, 24 and 36**. Hence the correct answer is (d).

39. $\text{LCM} = 2^2 \times 3^3 \times 5 \times 11 = 5940$

2	22,	54,	108,	135,	198
2	11,	27,	54,	135,	99
3	11,	27,	27,	135,	99
3	11,	9,	9,	45,	33
3	11,	3,	3,	15,	11
5	11,	1,	1,	5,	11
11	11,	1,	1,	1,	11
	1,	1,	1,	1,	1

40. $4 \times 27 \times 3125 = 2^2 \times 3^3 \times 5^5$
 $8 \times 9 \times 25 \times 7 = 2^3 \times 3^2 \times 5^2 \times 7$
 $16 \times 81 \times 5 \times 11 \times 49 = 2^4 \times 3^4 \times 5 \times 11 \times 7^2$
 $\therefore \text{HCF of above three numbers} = 2^2 \times 3^2 \times 5 = 180$

41. $\frac{1095}{1168} = \frac{15}{16}$

42. Let co-prime numbers be x, y and z .
 Given $xy = 551$ and $yz = 1073$.

$$\frac{yz}{xy} = \frac{1073}{551};$$

$$\begin{array}{r} 551 \overline{) 1073} \quad (1 \\ \underline{551} \\ 522 \overline{) 551} \quad (1 \\ \underline{522} \\ 29 \overline{) 522} \quad (18 \\ \underline{29} \\ 232 \\ \underline{232} \\ 0 \end{array}$$

HCF = 29

$$\frac{z}{x} = \frac{1073}{551} = \frac{37}{19}$$

and $y = \text{HCF} = 29$.

$$\therefore x + y + z = 19 + 29 + 37 = 85$$

43. $\text{LCM} = 48$; Ratio of numbers = $2 : 3$.

Let numbers be $2x$ and $3x$.

$$\therefore \text{LCM of numbers} = 2 \times 3 \times x = 48$$

$$\therefore x = 8$$

$$\therefore \text{Sum of numbers} = 2x + 3x = 5x = 5 \times 8 = 40$$

44. Given $\text{LCM of three numbers} = 2400$;

Ratio of numbers = $3 : 4 : 5$.

$$\begin{array}{r} 29 \overline{) 1073} \quad (37 \\ \underline{87} \\ 203 \\ \underline{203} \\ 0 \end{array}$$

$$\begin{array}{r} 29 \overline{) 551} \quad (19 \\ \underline{29} \\ 261 \\ \underline{261} \\ 0 \end{array}$$

Let numbers be $3x$, $4x$ and $5x$.

$$\therefore \text{LCM of numbers} = 3 \times 4 \times 5 \times x = 2400$$

$$\therefore x = \frac{2400}{60} = 40$$

i.e. HCF of numbers = 40

45. Given LCM of two prime numbers x and $y = 161$

$$\therefore \text{HCF of } x \text{ and } y = 1,$$

$$\text{LCM} = x \times y = 161;$$

Factorising, we get

$$x \times y = 161 = 7 \times 23$$

$$\therefore x = 23; y = 7$$

($\because x > y$)

$$\therefore 3y - x = 3 \times 7 - 23 = -2$$

46. $\therefore$ Difference between divisor and remainder = 10,

$$\text{Required number} + 10 = \text{LCM of } 48, 60, 72, 108 \text{ and } 140$$

$$\text{LCM} = 2^4 \times 3^3 \times 5 \times 7 = 15120$$

$$\therefore \text{Required number} = \text{LCM} - 10 = 15120 - 10 = 15110$$

2	48,	60,	72,	108,	140
2	24,	30,	36,	54,	70
2	12,	15,	18,	27,	35
2	6,	15,	9,	27,	35
3	3,	15,	9,	27,	35
3	1,	5,	3,	9,	35
5	1,	5,	1,	3,	35
	1,	1,	1,	3,	7

47. LCM of 4, 6, 10 and 15 = $2^2 \times 3^2 \times 5 = 60$

2	4,	6,	10,	15
5	2,	3,	5,	15
3	2,	3,	1,	3
	2,	1,	1,	1

Least number of six digits = 100000

Least number of six digits divisible by above 4 numbers

$$= 100000 - 40 + 60 = 100020$$

$$60) 100000 \text{ (1666)}$$

$$\begin{array}{r} 60 \\ \underline{60} \\ 400 \\ \underline{360} \\ 400 \\ \underline{360} \\ 400 \\ \underline{360} \\ 40 \end{array}$$

$\therefore$ Number leaves a remainder 2.

$\therefore$ Required number = $100020 + 2 = \mathbf{100022}$

$\therefore$ Sum of digits of number = $1 + 2 + 2 = 5$

48. Smallest number of five digits = 10000

LCM of 12, 16, 18 and 27

2	12,	16,	18,	27
2	6,	8,	9,	27
2	3,	4,	9,	27
3	3,	2,	9,	27
3	1,	2,	3,	9
	1,	2,	1,	3

$$= 2^4 \times 3^3 = \mathbf{432}$$

432) 10000 (23

$$\begin{array}{r} 864 \\ 1360 \\ 1296 \\ \hline 64 \end{array}$$

Smallest number of five digits exactly divisible by above numbers = $10000 - 64 + 432 = \mathbf{10368}$

49. Largest number of four digits = 9999

LCM of 14, 24, 27 and 32

2	14,	24,	27,	32
2	7,	12,	27,	16
2	7,	6,	27,	8
2	7,	3,	27,	4
3	7,	3,	27,	2
3	7,	1,	9,	2
	7,	1,	3,	2

$$= 2^5 \times 3^3 \times 7 = \mathbf{6048}$$

$\therefore$ Largest number of four digits exactly divisible by 14, 24, 27 and 32 = $\mathbf{6048}$

50. LCM of $2^2 \times 3^2 \times 5 \times 11$, $2^4 \times 3^4 \times 5^2 \times 7$ and $2^5 \times 3^2 \times 5^2 \times 7^2 \times 11$

$$= 2^2 \times 3^2 \times 5 = \mathbf{180}$$